🚲 Bike Dekho – Bike Sales Analysis

By

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for

Labmentix Internship 1st Project

Executive Summary

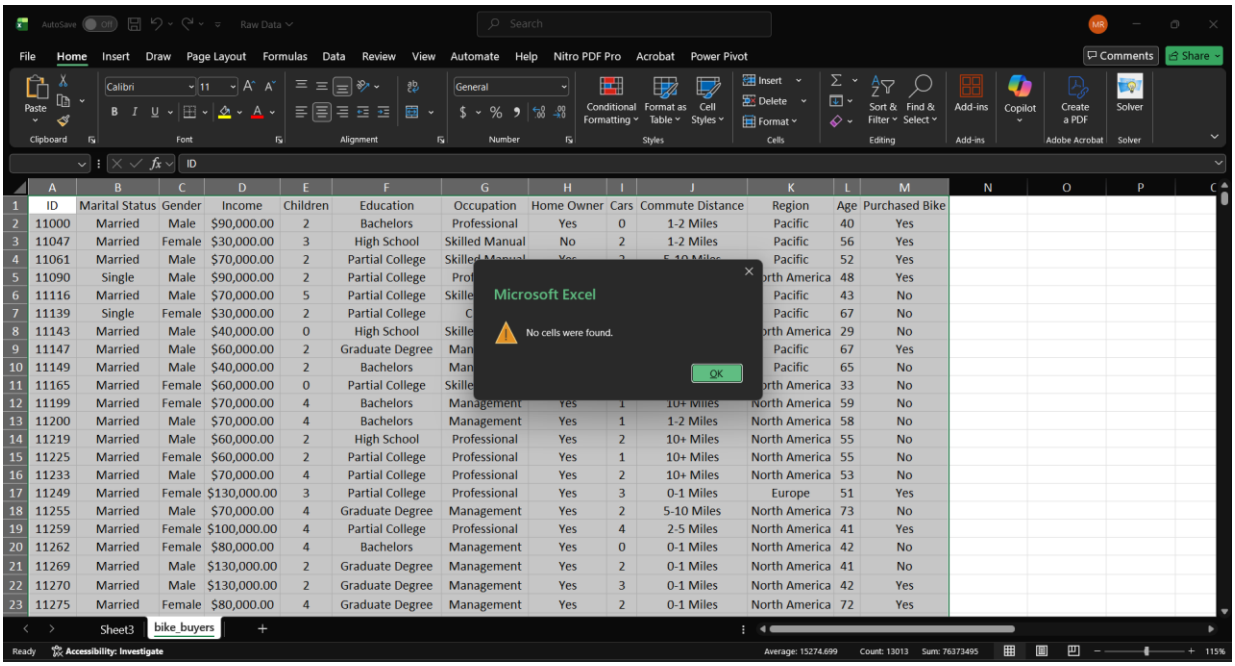
This project report shows an analysis of customer data based on demographic, socioeconomic, and behavioural patterns to identify key factors that influence bike purchases. Data cleaning, transformation, and exploratory data analysis were performed in Microsoft Excel using tools like pivot tables, visualization charts, formulas and a dashboard. Amongst the 1000 customers, an untapped market potential was found as only 48.10% were bike buyers, which indicates that the majority of them are non-buyers.

Some of the key insights that were derived from the data show that middle-aged, single bike buyers from the Pacific region are more numerous. Similarly, commute distance, education, and income were also seen as strong drivers, with customers travelling 2-5 miles being the most likely bike buyers, and buyers were mostly holders of a bachelor's degree, whereas in terms of income, buyers were seen more from the above-average income category. Household dynamics such as home ownership, car ownership, and the number of children also significantly influenced buying patterns. The findings provide actionable insights for targeted marketing, segmentation, and product positioning strategies.

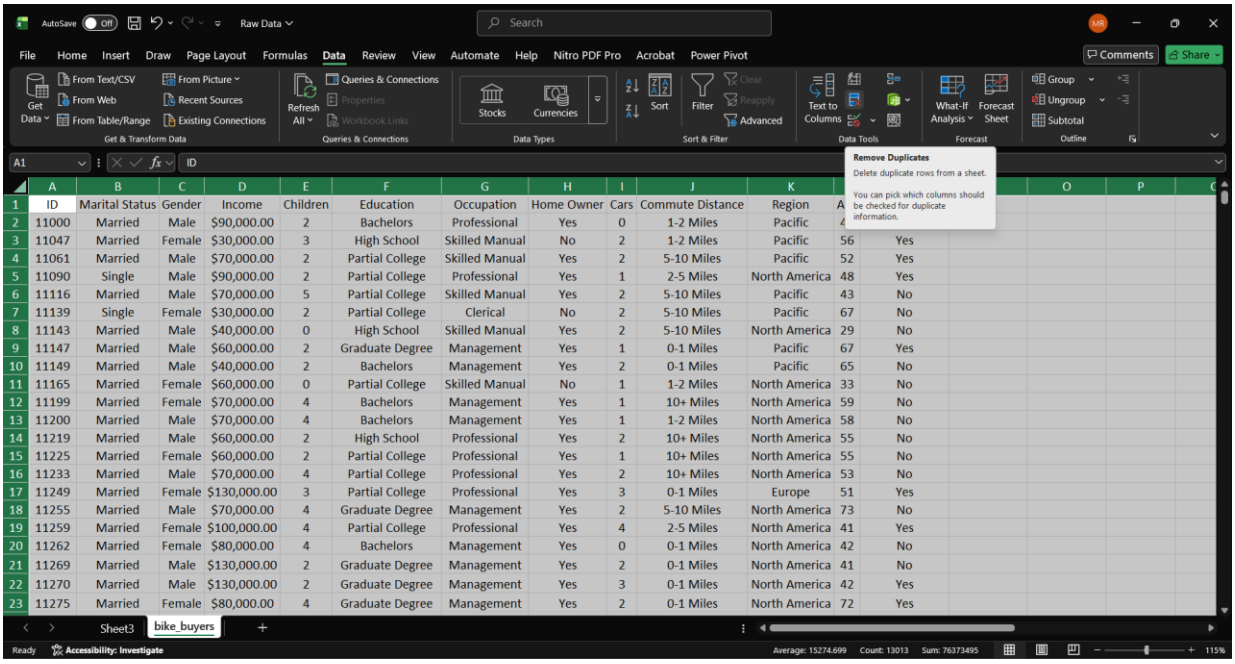
Key Tasks Performed

1. Data Cleaning

- The dataset was checked for Nulls using the Go to special option:



- Duplicates were removed using the remove duplicates option:



- Using the find and replace option, the data under columns “Marital Status” and “Gender” were changed from “M” and “S” to “Married” and “Single”. Similarly, under column “Gender,” the data were changed from “F” and “M” to “Female” and “Male” for ease of interpretation.
- After calculating the **average income ($56,360)** from column D using the average formula (**=AVERAGE(D:D)**), a new column was created using the IF function (**=IF(D2>56360, "Yes", "No")**) and was named “Above Average Income”. A second column was created with the name “Own Car” using the IF function (**=IF(J2>0, "Yes", "No")**) and the data from the “Cars” column (J column) to segregate customers who had owned a car from customers who did not own one.
- Similarly, the column “Age” was categorized in a new column (Age Group) using the IF function (**=IF(N2<30, "Young Adult", IF(N2<50, "Middle Aged", "Older Adult")**) to categorize cutomers based on their age into 3 categories namely Young Adult, Middle Aged, and Older Adult.

2. **Exploratory Data Analysis and insights using data visualizations:**

- Customer base insights using pivot tables;

Customer Base

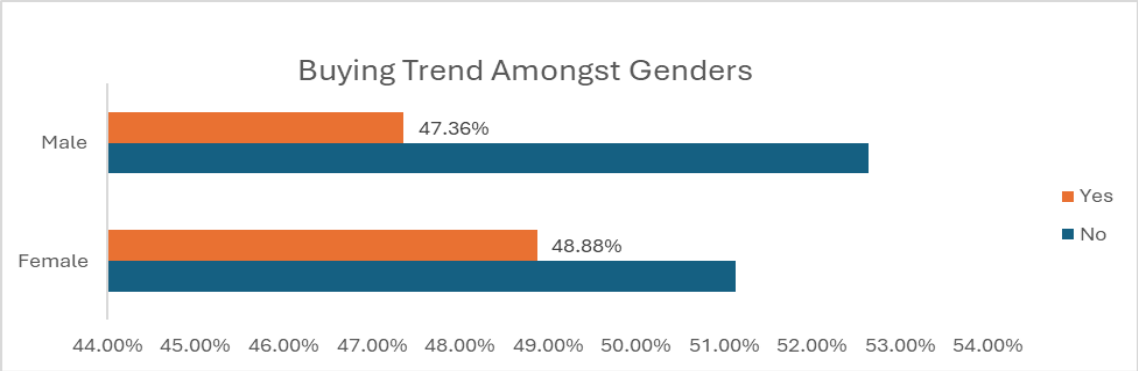
Row Labels	Count of ID
No	51.90%
Yes	48.10%
Grand Total	**100.00%**

Count of Bike Buyers (by Gender)

Count of ID	Column Labels		
Row Labels	Female	Male	Grand Total
⊟ Yes	239	242	481
⊞ Middle Aged	167	157	324
⊞ Older Adult	57	65	122
⊞ Young Adult	15	20	35
Grand Total	**239**	**242**	**481**

Total Number of Customers that bought a bike (by Gender)

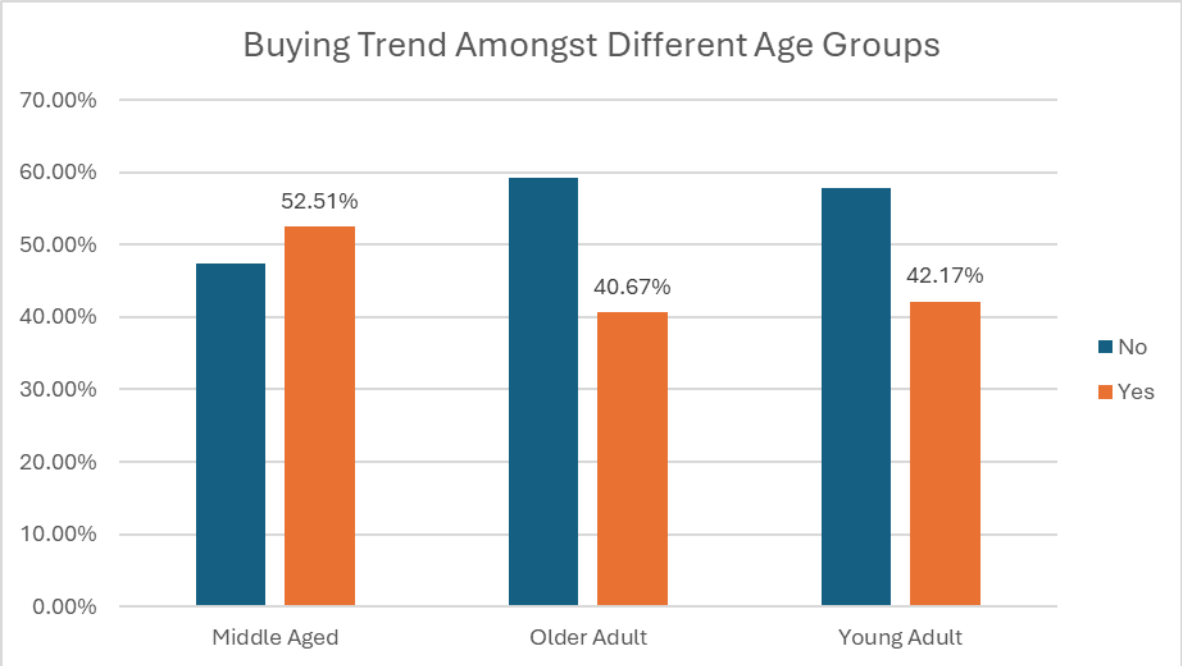
Count of ID	Column Labels		
Row Labels	No	Yes	Grand Total
Female	51.12%	48.88%	100.00%
Male	52.64%	47.36%	100.00%
Grand Total	**51.90%**	**48.10%**	**100.00%**



The actual customer base (people who have bought a bike) was found to be 48.10%, which revealed that 51.90% people are non-buyers. Even though the number of male buyers is more (242) and female buyers are 239, the conversion rate of female customers into buyers is seen to be slightly more, as female customers are 48.88% likely to buy a bike, whereas male customers have a slightly reduced percentage of 47.36% converstion rate of becoming a buyer.

- Buying Trend Amongst Different Age Groups

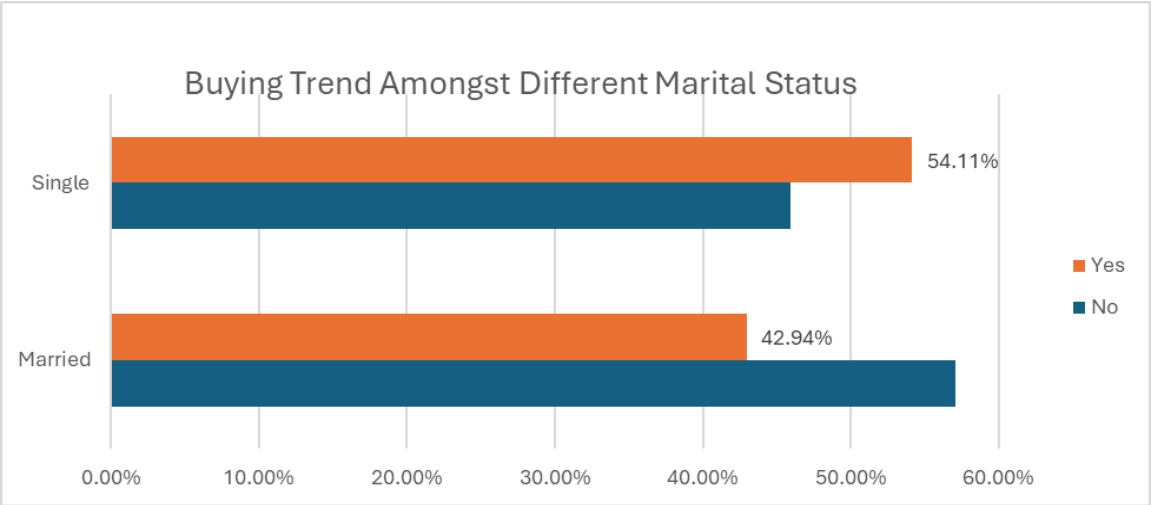
Buying Trend Amongst Different Age Groups			
Count of ID	Column Labels		
Row Labels	No	Yes	Grand Total
Middle Aged	**47.49%**	**52.51%**	**100.00%**
Older Adult	59.33%	40.67%	100.00%
Young Adult	57.83%	42.17%	100.00%
Grand Total	**51.90%**	**48.10%**	**100.00%**



Buying trend amongst different age groups shows that middle-aged buyers are more in numbers, with 52.51%, followed by young adults with 42.17%. The least number of buyers was found in the older adult age group, with 40.67%.

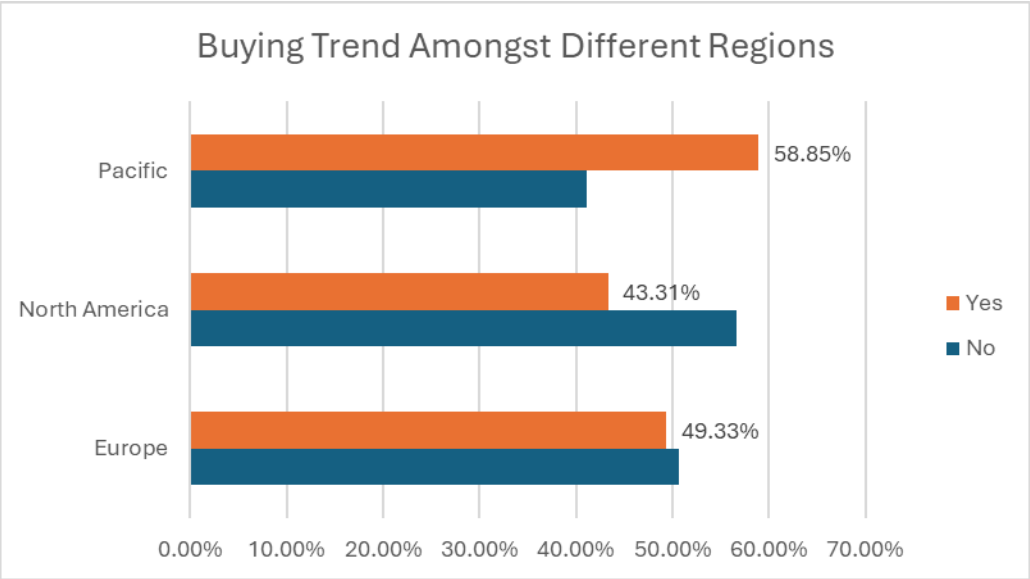
- Buying Trend Amongst Different Marital Status

Buying Trend Amongst Different Marital Status			
Count of ID	Column Labels		
Row Labels	No	Yes	Grand Total
Married	57.06%	42.94%	100.00%
Single	**45.89%**	**54.11%**	**100.00%**
Grand Total	**51.90%**	**48.10%**	**100.00%**



Amongst Different marital statuses, the most buyers were found to be single, with 54.11,% while married buyers were 42.94%.

- Buying Trend Amongst Different Regions

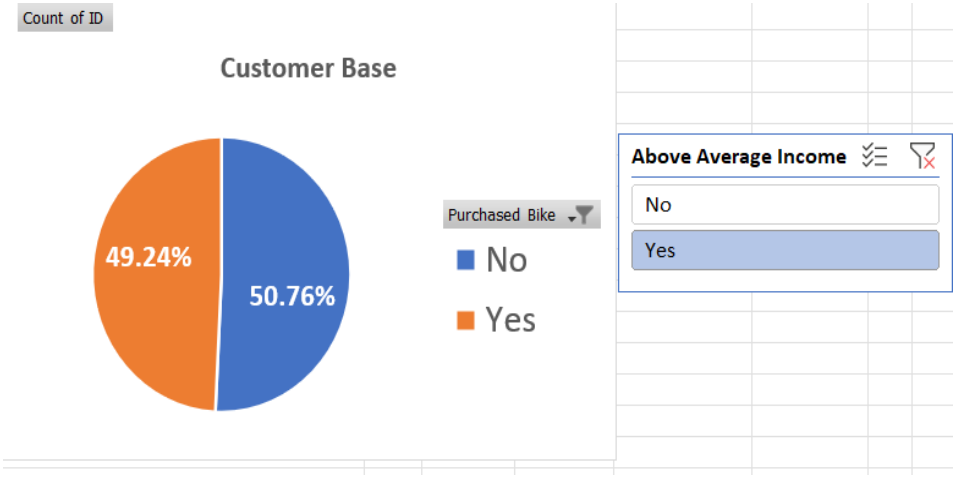


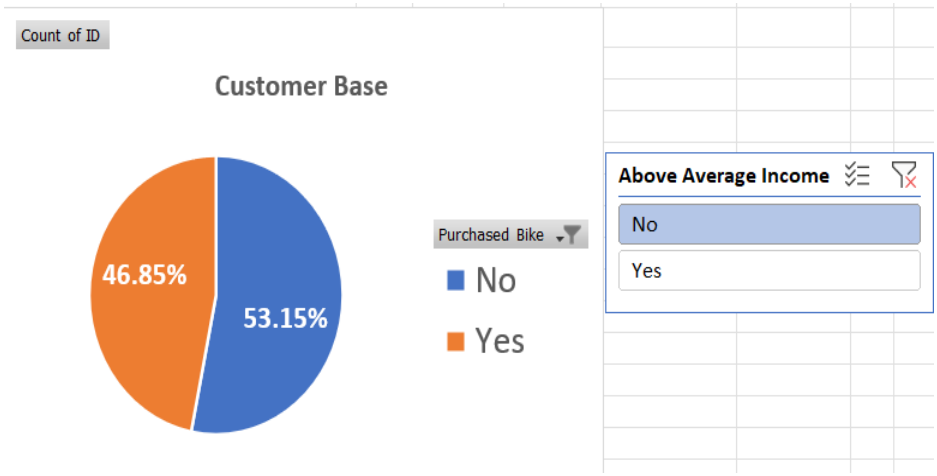
Buying Trend Amongst Different Regions

Count of ID	Column Labels		
Row Labels	**No**	**Yes**	**Grand Total**
Europe	50.67%	49.33%	100.00%
North America	56.69%	43.31%	100.00%
Pacific	**41.15%**	**58.85%**	**100.00%**
Grand Total	**51.90%**	**48.10%**	**100.00%**

Analyzing the buying trend amongst different regions revealed that the Pacific region has a higher buyer rate of 58.85%, followed by Europe with 49.33%. The lowest buyer rate was found to be in the North American region, with 43.31% buyers.

- Customer base based on income





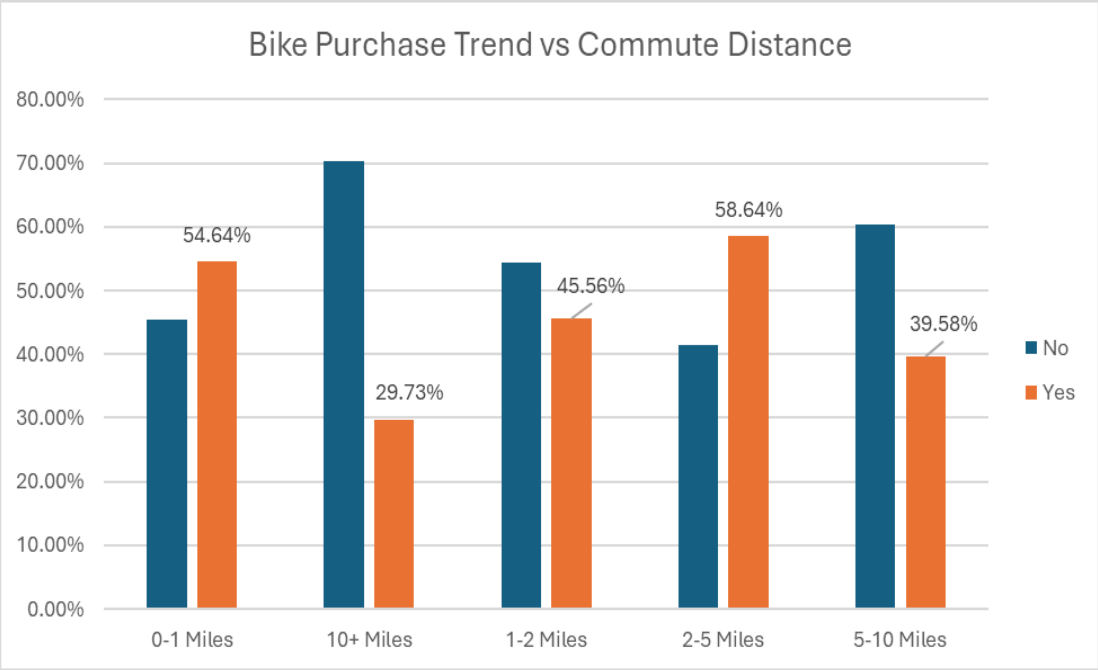
Income based Insight

Count of ID	Column Labels	
Row Labels	Yes	Grand Total
No	223	223
Yes	258	258
Grand Total	**481**	**481**

The number of customers with above average income is 258, while buyers with below average income were found to be 223. Customers who are bike buyers are fewer in both categories (above average income = 49.24% and below average income = 46.85%).

- Bike Purchase Trend vs Commute Distance

Bike Purchase Trend With Respect to Commute Distance			
Count of ID	Column Labels		
Row Labels	No	Yes	Grand Total
0-1 Miles	45.36%	54.64%	100.00%
10+ Miles	70.27%	29.73%	100.00%
1-2 Miles	54.44%	45.56%	100.00%
2-5 Miles	**41.36%**	**58.64%**	**100.00%**
5-10 Miles	60.42%	39.58%	100.00%
Grand Total	**51.90%**	**48.10%**	**100.00%**



Analyzing the bike purchase data with the commute distance found that 58.64% (majority) bike buyers commute a distance of 2-5 miles, followed by 54.64% buyers commuting a distance of 0-1 miles. Commute distances more than 5 miles see a decline from 39.58% (5-10 miles) to 29.73% (10+ miles).

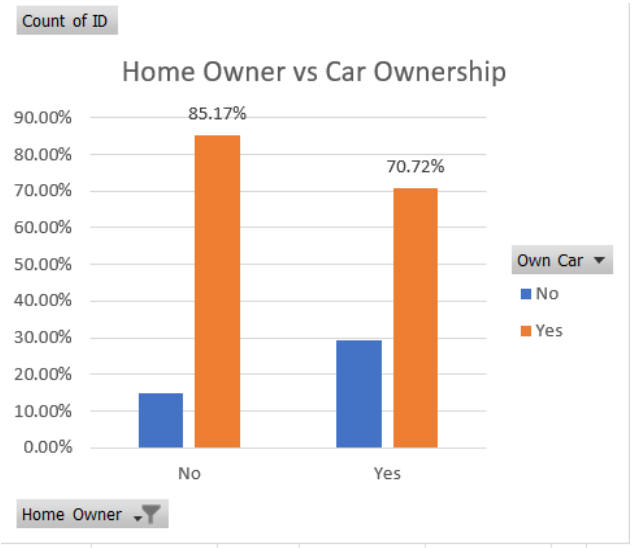
- Comparitive study of Home Owner vs Car Ownership and Home Owner vs Bike Purchase

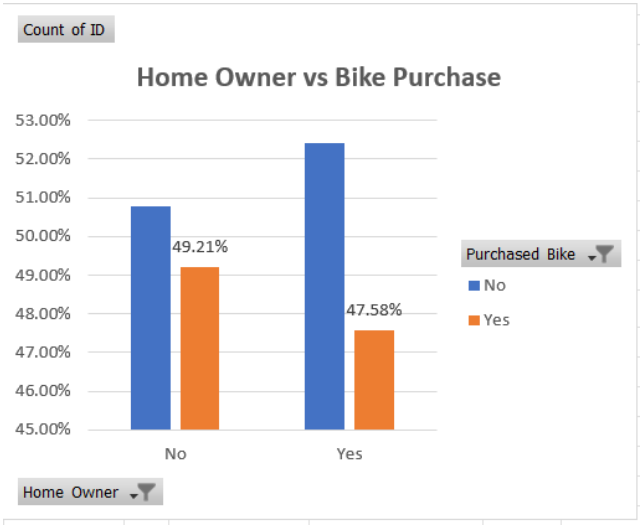
Home Owner vs Car Ownership

Count of ID	Column Labels		
Row Labels	No	Yes	Grand Total
No	**14.83%**	**85.17%**	**100.00%**
Yes	29.28%	70.72%	100.00%
Grand Total	**24.70%**	**75.30%**	**100.00%**

Home Owner vs Bike Purchase

Count of ID	Column Labels		
Row Labels	No	Yes	Grand Total
No	**50.79%**	**49.21%**	**100.00%**
Yes	52.42%	47.58%	100.00%
Grand Total	**51.90%**	**48.10%**	**100.00%**



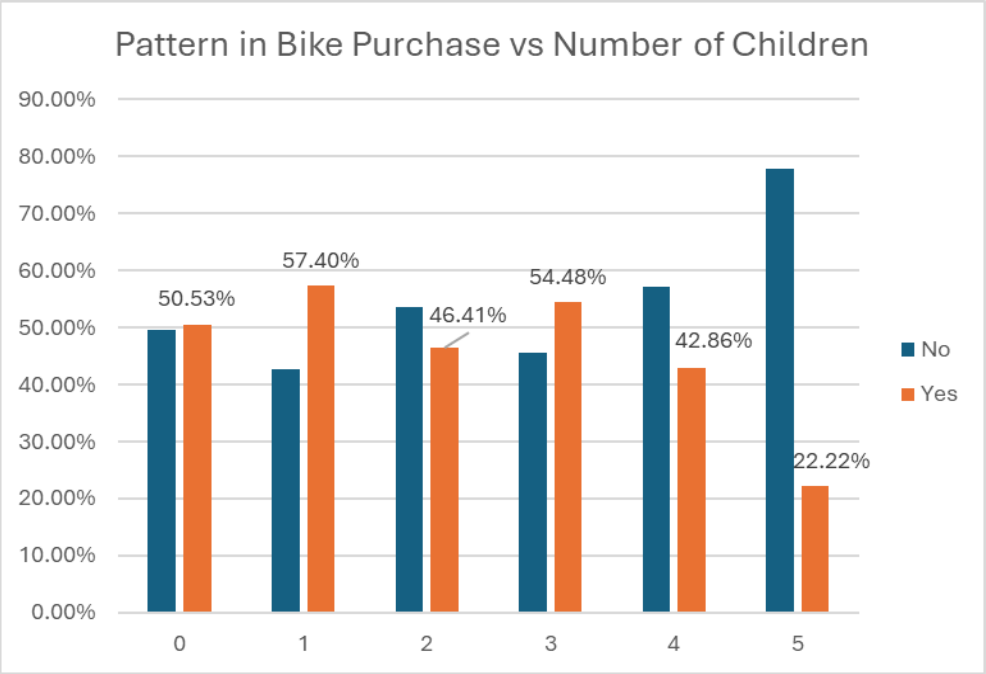


Comparing these 2 data, we see that homeowners often have cars and do not buy bikes. The data reveals that buyers are not homeowners 49.21% of the time, whereas homeowners buy a bike 47.58% of the time. This could be because of the fact that 70.72% homeowners own a car and do not need a bike.

- Pattern in Bike Purchase vs Number of Children

Pattern in Bike Purchase vs Number of Children

Count of ID	Column Labels		
Row Labels	No	Yes	Grand Total
0	49.47%	50.53%	100.00%
1	42.60%	57.40%	100.00%
2	53.59%	46.41%	100.00%
3	45.52%	54.48%	100.00%
4	57.14%	42.86%	100.00%
5	**77.78%**	**22.22%**	**100.00%**
Grand Total	**51.90%**	**48.10%**	**100.00%**

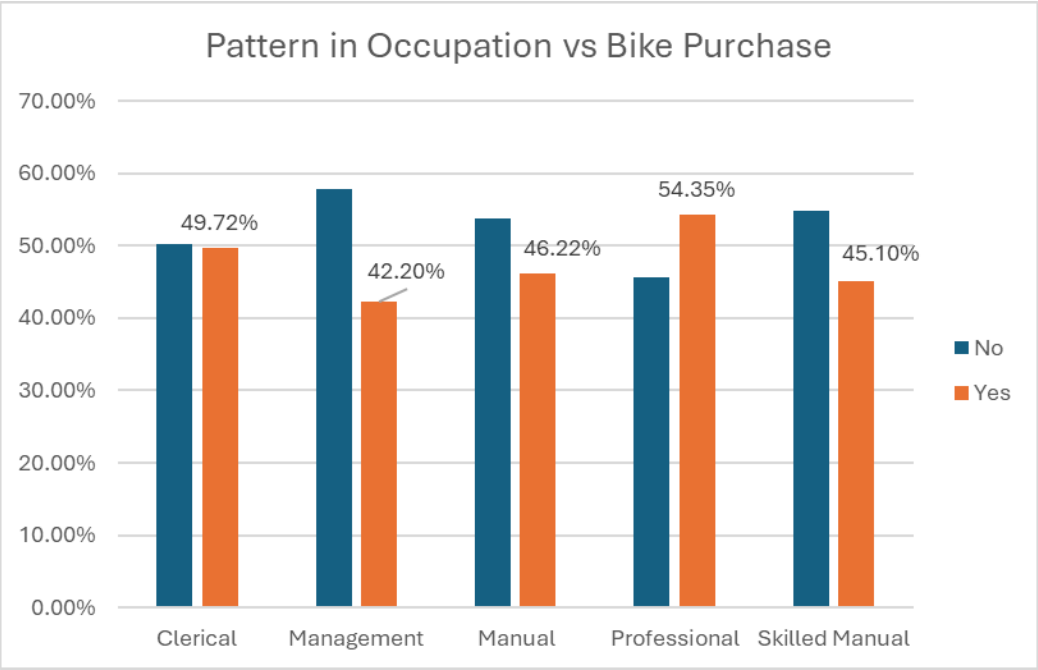


This comparison between bike purchase and number of children reveals that families with 3 or more children do not prefer buying a bike, as the number of bike purchases drops from 54.48% (3 children) to 22.22% (5 children). Most buyers have 1 child (57.40%), 3 children (54.48%), or none (50.53%).

- Pattern in Occupation vs Bike Purchase

Pattern in Occupation vs Bike Purchase

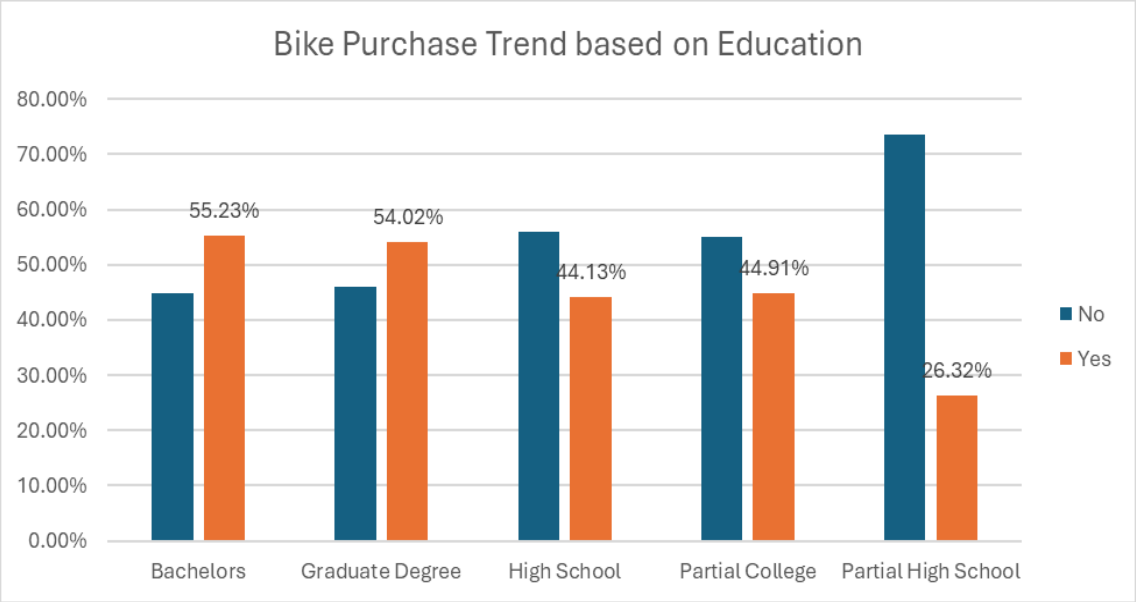
Count of ID	Column Labels		
Row Labels	**No**	**Yes**	**Grand Total**
Clerical	50.28%	49.72%	100.00%
Management	57.80%	42.20%	100.00%
Manual	53.78%	46.22%	100.00%
Professional	**45.65%**	**54.35%**	**100.00%**
Skilled Manual	54.90%	45.10%	100.00%
Grand Total	**51.90%**	**48.10%**	**100.00%**



A trend in occupation amongst bike buyers is observed, which suggests that bike buyers are often from the occupational category of professionals (54.35%), followed by clerical (49.72%), and Manual (46.22%). The least buyers are from the management (42.20%) occupational category.

- Bike Purchase Trend based on Education

Count of ID	Column Labels		
Row Labels	No	Yes	Grand Total
Bachelors	**44.77%**	**55.23%**	**100.00%**
Graduate Degr	45.98%	54.02%	100.00%
High School	55.87%	44.13%	100.00%
Partial College	55.09%	44.91%	100.00%
Partial High Sch	73.68%	26.32%	100.00%
Grand Total	**51.90%**	**48.10%**	**100.00%**



Amongst buyers from different educational backgrounds, people with bachelor's degrees and graduate degrees were seen to be the most common.

3. Business Recommendations

- Targeting middle-aged customers with strategies focusing on commute distances (5 miles and below), occupation (professionals and clerical), and educational background (bachelor's and graduate degree holders).
- Expanding women-centric product designs and focusing on marketing strategies for the same.
- Increasing presence in high-performing regions like the Pacific and trying to replicate the success strategy in other regions.
- Providing financing options targeting non-homeowners and buyers of lower income category.
- Partnering with companies based on the educational and occupational backgrounds of buyers.
- Introducing family-friendly bike accessories that could convince households with more than 1 children to consider buying a bike.
- Positioning bikes as a recreational tool while marketing them to car owners.

4. Conclusion

The analysis shows that bike purchasing patterns are driven by daily commuting needs and lifestyle rather than by income alone. Targeted segmentation, regional strategies, and occupation-based marketing can increase the number of customers that turn into buyers. These insights provide the necessary steps and direction for the business’s decision-making and future expansion strategies.